DIGI Node — Tier 2: Complete Parts Kit

The kit has every part you need to build a finished node, as the suppliers ship them. Nothing is soldered and nothing is flashed.

In the box

Group	Contents
Electronics	Bare 82 mm PCB, all modules and headers, 8 GB microSD
Enclosure	Printed enclosure, TPU seals, all hardware and wiring
RF	Wi-Fi and LoRa SMA antennas, U.FL pigtails and bulkheads, passive ceramic GPS antenna
Power and cooling	Fan, thermal switch, power switch, waterproof USB-C panel jack, UPS charger board, JST terminals and cables

You supply: two flat-top 18650 batteries, an active GPS SMA antenna and soldering tools.

Build it

1. Follow the assembly manual from start to finish. It has photographed steps for soldering, connectors, the enclosure, the SD card and power wiring. **github.com/lukeswitz/AntiHunter → `hw` folder → `Antihunter-DIGINODE-AssemblyManual.pdf`**
2. Two modules need their headers modified first (manual, page 6):
   - The RTC comes with female headers.
   - Some tamper sensors come with inverted 90° headers.
3. Fit the SD card before you close the case. You can't reach the slot once the PCB is mounted.

Flash it

4. Flash the Heltec V3.2 or T114 with the latest stable Meshtastic from flasher.meshtastic.org, then set up the serial module as below. Without these settings the node and the radio cannot communicate. `scripts/meshtastic_config.py` in the repo applies them for you.

Setting	Heltec V3	T114
Serial module	Enabled	Enabled
Mode	TEXTMSG	TEXTMSG
Baud	115200	115200
RX / TX pins	19 / 20	10 / 9

5. Flash AntiHunter on the ESP32 over USB with the web flasher at lukeswitz.github.io/AntiHunter. Choose **Full** for the web UI or **Headless** for quiet field operation.
6. Set the LoRa region and change the Bluetooth pairing PIN. Make your own encrypted channel the primary channel and turn the public default off.
7. **Full firmware only:** join the `Antihunter` Wi-Fi network with the password `antihunt3r123`, open `http://192.168.4.1` and change these credentials first.
8. Write down the erase key. The node makes one on first boot. Press **Show my erase key** in the web flasher, or read ERASE KEY in the serial log. Change it in the web UI under System → Settings → **Danger zone**, or over mesh with `CONFIG_ERASE_PSK`.

Power

- Use a regulated 5 V supply only, from USB or the UPS board. Never wire cells or an unregulated supply to the PCB.
- A 2 A fast-blow fuse in the battery line is optional. It is not included.

> **Connect one USB cable at a time.** The ESP32 and the radio share the power rails.

Full guide

Operator's Guide: github.com/lukeswitz/AntiHunter/blob/main/docs/AntiHunter-Operators-Guide.pdf

- **Firmware, docs, issues:** github.com/lukeswitz/AntiHunter
- **Command Center:** github.com/TheRealSirHaXalot/AntiHunter-Command-Control-PRO